

Exact Physically Assisted Quadrature of the Circle by the Hydrogeometric Transfer Method

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Abstract

Squaring the circle is a classical problem in geometry, first proposed in ancient Greek mathematics. It involves constructing a square with the same area as a given circle using only a finite number of steps with a compass and straightedge. In 1882, this task was proven to be impossible as a consequence of the Lindemann–Weierstrass theorem, which establishes that pi is a transcendental number—that is, it is not a root of any polynomial with rational coefficients. At first glance, the problem appears very simple, and its visual nature makes it understandable even to an elementary school student. This is why it has remained so popular and continues to attract the interest of many mathematicians. Despite the proof of its impossibility, attempts to square the circle have been common in mathematical crankery. As a result, numerous solutions with varying degrees of approximation accuracy have been proposed, along with approaches that rely on additional tools—one of which is the method described here.

1 Introduction

Given a circle of radius r , one seeks a square of equal area. Since such a square must have side length $l = r\sqrt{\pi}$, the problem is equivalent to constructing the length $r\sqrt{\pi}$ from the given radius r .

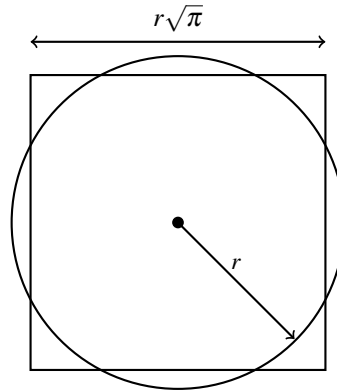


Figure 1: A circle of radius r and a square of side $r\sqrt{\pi}$ have the same area πr^2 .

The present work is situated in an enlarged framework of admissible operations. Alongside the usual straightedge and compass, the construction admits two vessels containing water, both of equal depth h , whose relevant dimensions can themselves be specified by Euclidean construction. One vessel is a cylinder of radius r , and the other is a right isosceles triangular prism whose legs have length $2r$.

The purpose of this extension is to convert an area relation into a measurable length relation by means of volume conservation. The Hydrogeometric Transfer Method uses an exact transfer of water between the two vessels to determine a length $l = r\sqrt{\pi}$, from which a square equal in area to the given circle is obtained.

2 The Hydrogeometric Transfer Method

The method rests on a single simple idea: conservation of volume converts the area of a circle into a measurable length inside a triangle. The construction uses two vessels of equal depth.

The Two Aquariums

Aquarium 1 — Cylindrical aquarium. See Figure 2. The cylinder is oriented so that its circular cross-section faces the viewer directly: the front face is a circle of radius r and the back face is an identical circle at depth h . The axis of the cylinder is therefore horizontal, pointing away from the viewer. Let r be the radius of that circle and h the depth of the aquarium. The area of the circular base is

$$A = \pi r^2,$$

and the volume of the full cylinder is

$$V = A \cdot h.$$

Aquarium 2 — Triangular prism. See Figure 2. The prism is oriented so that its triangular cross-section faces the viewer directly. The base is a right isosceles triangle with the hypotenuse at the top and in a strictly horizontal position; the right-angle vertex points downward. Each leg of the triangle equals the diameter $2r$ of the circle, and the depth of the prism equals the same h as the cylindrical aquarium. The area of the triangular base is

$$A_{\triangle} = \frac{1}{2}(2r)^2 = 2r^2,$$

and the volume of the full prism is

$$V_{\triangle} = A_{\triangle} \cdot h = 2r^2 h.$$

All dimensions required to build these two vessels and place them as described can be constructed exactly using only a straightedge and compass.

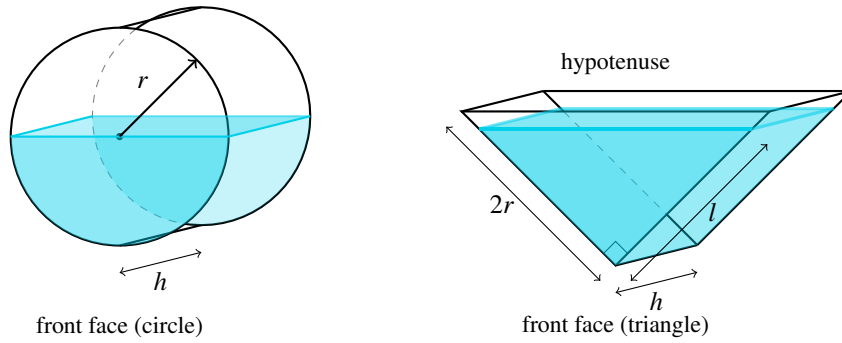


Figure 2: Left: cylindrical aquarium (front view is a circle of radius r , depth h). Right: triangular-prism aquarium (front view is a right isosceles triangle with hypotenuse on top, legs $= 2r$, depth h).

The Transfer Argument

Fill the cylindrical aquarium to exactly half its capacity. The water occupies a half-cylinder whose volume is

$$V_{\text{water}} = \frac{1}{2} A_{\text{circle}} \cdot h = \frac{1}{2} \pi r^2 h.$$

Now pour all of this water into the triangular prism. Because both containers share the same depth h , the water level in the prism forms a triangle on the front face of the prism whose area equals

$$A_{\text{front triangle water}} = \frac{V_{\text{water}}}{h} = \frac{1}{2} \pi r^2.$$

The key observation is:

$$A_{\text{front triangle water}} = \frac{1}{2} A_{\text{circle}}.$$

Connecting Triangle to Circle

The procedure above shows that the area of the right isosceles triangle whose legs have a specific length l satisfies

$$\frac{1}{2} l^2 = \frac{1}{2} \pi r^2 \implies l = r\sqrt{\pi}.$$

In other words, if the right isosceles triangle is constructed with legs $l = r\sqrt{\pi}$, its area equals $\frac{1}{2} \pi r^2$.

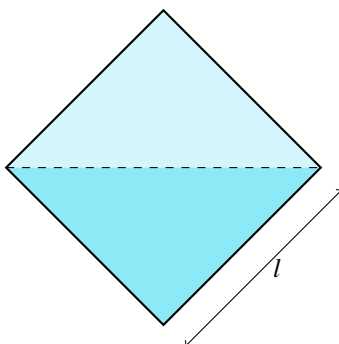


Figure 3: The water triangle and its reflection across the hypotenuse form a square of side l .

Reflecting that triangle across its hypotenuse, as in Figure 3, gives a square with side $l = r\sqrt{\pi}$ and area

$$l^2 = \pi r^2 = A_{\text{circle}},$$

which is exactly the area of the original circle.

3 Conclusions

The Hydrogeometric Transfer Method has been presented as a physically assisted extension of geometric construction for the quadrature of the circle. The method does not contradict the classical impossibility of straightedge-and-compass quadrature; rather, it operates in an enlarged framework in which two Euclidean-constructible vessels and an exact transfer of water are admitted as additional resources.

Within this framework, conservation of volume produces the length $l = r\sqrt{\pi}$ from a given radius r , and this length in turn determines a square of area πr^2 . The significance of the result lies in the fact that even if the numerical value of π were not known, the area of the circle could still be treated as a geometric magnitude that passes into a different form. In the present construction, that same area becomes visible as a length, namely the side of a square, and therefore one may assert directly that the area of the given circle is equal to the area of the constructed square.

References

- [1] F. Lindemann, *Ueber die Zahl π* , *Mathematische Annalen* **20** (1882), 213–225. doi:10.1007/BF01446522
- [2] E. W. Hobson, *Squaring the Circle: A History of the Problem*, Cambridge University Press, Cambridge, 1913.